LaBTECH			Title:	**Checking of Plate Load Test Setup (Max. 750kPa)**
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Design Assumption

Unit weight of Concrete Block	=	24	kN/m^3
(CoPFD-Cl.2.2.2(5)) Presumed allowable bearing pressure on ground	=	100	kPa

Setup Information

Test pressure at 3W	=	750	kPa
Plate width	=	0.3	m (rectangular)
Plate area	=	0.09	m^2
Test load	=	67.5	kN
Size of concrete block	=	750 x 750 x 2250	
Self-weight of each concrete block	=	30.38	kN
Size of beam SB1	=	UC 305x305x97kg/m	
Length of SB1	=	5.00	m
Self-weight of each SB1	=	4.76	kN
Size of beam MB1	=	UC 305x305x97kg/m	
Length of MB1	=	2.25	m
Self-weight of each MB1	=	2.14	kN

Check: Bearing Pressure of Supporting Concrete Block

Loading

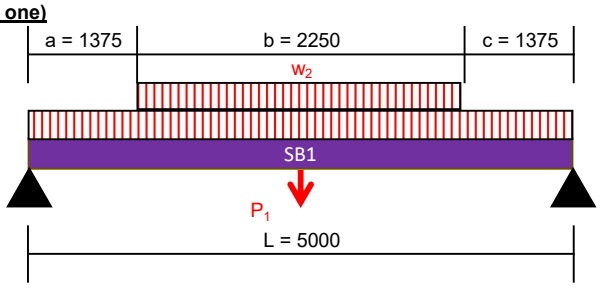
3 nos of concrete block on top	=	91.125	kN
3 nos of SB1	=	14.274	kN
1 nos of MB1	=	2.141	kN
2 nos of supporting concrete block	=	60.750	kN
Total Dead Load (DL)	=	168.290	kN

Base area of each supporting concrete block, A = 0.75 x 2.25 = 1.6875 m^2

Bearing pressure of supporting concrete block = 168.29 / 2nos / 1.69 = 49.9 kPa < 100 kPa OK

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Check: SB1 (middle one)



UC 305x305x97kg/m
S355

Loading

P_1 = Self-weight of MB1
= 1.4 x 2.14
= 3.00 kN (factored)

w_1 = Self-weight of SB1
= 1.4 x 97 x 9.81/1000
= 1.33 kN/m (factored)

w_2 = Self-weight of concrete block on top
= 1.4 x 24 x 0.75 x 1.125
= 28.35 kN/m (factored)

Design Shear, V = $P_1/2 + w_1L/2 + w_2b/L\ (b/2 + c)$
= 3 / 2 + 1.33 x 5 / 2 + 28.35 x 2.25 / 5 x (2.25/2 + 1.375)
= 36.72 kN

Design Moment, M = $P_1L/4 + w_1L^2/8 + w_2\ [(L/2)^2 - a^2]/2$
= 3 x 5 / 4 + 1.33 x 5^2 / 8 + 28.35 x [(5/2)^2 - 1.375^2]/2
= 69.70 kNm

Structural Checking

(STRUCTURAL CHECKING REFER NEXT PAGES)

Welding Checking

Weld Length, Lw = 305.3 x 4 nos
= 1221.2 mm

Weld Size, S = 6 mm F.W.

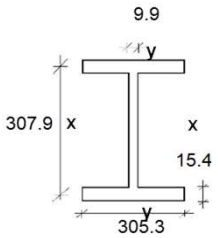
Weld Strength, pw = 220 MPa

Weld Capacity = 0.707 x S x pw x Lw /1000
= 0.707 x 6 x 220 x 1221 /1000
= 1140 kN
> 3.00 kN OK

Calculation Sheet	Job No.	Sheet No.	Rev.
			P.
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Job Tilte :	Drg. Ref.:		
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DESIGN CODE

Code of Practice for the Structural Use of Steel 2011



MEMBER SPECIFICATION

Member

Steel grade	=	BS S355	
Section	=	UC 305x305x97	(H-section)
Type	=	hot-rolled steel section	

MATERIAL/SECTIONAL PROPERTIES

Modulus of elasticity	E	=	205000	MPa								
Mass	m	=	96.9	kg/m								
Area	A	=	123	cm^2								
Dimensions	D	=	307.9	mm	d	=	246.7	mm				
	B	=	305.3	mm	b	=	152.65	mm				
	T	=	15.4	mm	t	=	9.9	mm				
Moment of inertia	Ix	=	22250	cm^4	Iy	=	7308	cm^4	Iu	= N.A.	cm^4	Iv = N.A. cm^4
Radius of gyration	rx	=	13.4	cm	ry	=	7.69	cm	ru	= N.A.	cm	rv = N.A. cm
Elastic modulus	Zx	=	1445	cm^3	Zy	=	479	cm^3				
Plastic modulus	Sx	=	1592	cm^3	Sy	=	726	cm^3				
	Svx	=	150.6	cm^3	Svy	=	717.7					
Torsional modulus	J	=	91.2	cm^4								
Design strength	py	=	355	MPa								

SECTION CLASSIFICATION

Parameter	ε	=	SQRT(275 / py)		=	0.8801		
Stress ratio	r1	=	Nc / (d * t * py)		=	0		
	r2	=	Nc / (A * py)		=	0		
Dimension ratio	b/T	=	9.9	<= 15 ε	=	13.2	Flange:	Class 3
	d/t	=	24.9	<= 80 ε / (1+r1)	=	70.4	Web:	Class 1
							Overall:	Class 3

DESIGN FORCES AND MOMENTS

Nc (kN)	Nt (kN)	Vx (kN) [along]	Vy (kN) [along]	Mx (kNm) [abt]	My (kNm) [abt]	Tr (kNm)
0.00	0.00	0.00	36.72	69.70	0.00	0.00

SHEAR CAPACITY CHECK

Shear area	Avy	= t * D	=	3048	mm^2					*(Cl. 8.2.1)*	
Shear capacity	Vcy	= py * Avy / SQRT(3)	=	624.76	kN	>=	Vy =	36.72	kN	OK	6%
Shear load check	Vy	<= 0.6 * Vcy	=	374.86	kN		(Low shear case)				
Shear area	Avx	= 0.9 * (2 * B * T)	=	8463	mm^2						
Shear capacity	Vcx	= py * Avx / SQRT(3)	=	1734.6	kN	>=	Vx =	0	kN	OK	0%
Shear load check	Vx	<= 0.6 * Vcx	=	1040.7	kN		(Low shear case)				

MOMENT CAPACITY CHECK

(Cl. 8.2.2)

(Low Shear Case)

Moment capacity	Mcx	= py * Zx	=	512.98	kNm	>=	Mx =	69.7	kNm	OK	14%
Moment capacity	Mcy	= py * Zy	=	170.05	kNm	>=	My =	0	kNm	OK	0%

BIAXIAL MOMENTS: LOCAL CAPACITY CHECK

$$\frac{Mx}{Mcx} + \frac{My}{Mcy} = \frac{69.7}{512.98} + \frac{0}{170.045} = 0.14 \quad <= \quad 1$$

OK 14%

Calculation Sheet	Job No.	Sheet No.	Rev.
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TENSION CAPACITY CHECK

Effective Area $a_e = Ke * a_n$ = 1.1 x(12300 (Ag) - 0 (Area deduction for hole in mm^2)) *(Cl. 8.6.1)*

= 13530 mm^2 >= 12300 mm^2 (Therefore take Ag)

Connection Type: **Welded Connection**

$a_2 = Ag - a_1$ = 9251.8 mm^2

Tension capacity $Pt = py * (a_e - 0 \; a_2)$ = 4366.5 kN >= Nt = 0 kN **OK** 0%

COMPRESSION CAPACITY CHECK *(Cl. 8.7.5)*

Effective length	Lcx =	**5.00** m	Lcy =	**5.00** m	
Slenderness	λx = Lcx / rx		λy = Lcy / ry		
	= 37.3134		= 65.0195		
From Table 8.7, use buckling curve	[b]		[c]		
Robertson constant	α = 3.5		α = 5.5		
	$pEx = (\pi^2 * E) / \lambda x^2$		$pEy = (\pi^2 * E) / \lambda y^2$		
	= 1453		= 479		
Limiting slenderness	λo = 0.2 * SQRT(π^2 * E / py)				
	= 15.0988				
Perry factor	η = α * (λx - λ0) / 1000		η = α * (λy - λ0) / 1000		
	= 0.078		= 0.275		
	ϕc = [py + (η + 1) * pEx] / 2		ϕc = [py + (η + 1) * pEy] / 2		
	= 961		= 482		
Compressive buckling strength	pcx = pEx * py / [ϕc + SQRT(ϕc^2 - pEx * py)]		pcy = pEy * py / [ϕc + SQRT(ϕc^2 - pEy * py)]		
	= 322.742 MPa		= 231.691 MPa		
Compression resistance	Pc = A * pcy = 2849.8 kN	>= Nc = 0 kN		**OK**	0%

LATERAL-TORSIONAL BUCKLING CHECK *(Cl. 8.3.1)*

Effective length	L_{LT} =	**5.00** m	m_{LT} =	**1**	
Slenderness	λ = L_{LT} / ry		= 65.0195		
Dist. betw. flange shear centers	hs = D - T		= 292.5 mm		
Web depth	hw = D - 2 * T		= 277.1 mm		
wraping constant for channel	H =		= 1.9E+12 mm^6		
Torsional index	x = 0.566 * hs * SQRT(A / J)		= 19.2264		
	γ = 1 - Iy / Ix		= 0.67155		
Buckling parameter	u = $(4 * Sx^2 * \gamma)/(A^2 * hs^2)]^{0.25}$		= 0.85161		
Slenderness factor	v = $1 / [1+ 0.05 * (\lambda / x)^2]^{0.25}$		= 0.8931		
	βw = Zx / Sx		= 0.90766		
Equivalent slenderness	λ_{LT} = u * v * λ * SQRT(βw)		= 47.1135		
	pE = $(\pi^2 * E) / \lambda_{LT}^2$		= 911.511 MPa		
Limiting equivalent slenderness	λ_{L0} = 0.4 * SQRT(π^2 * E / py)		= 30.1976		
Perry coefficient	η_{LT} = (λ_{LT} - λ_{L0}) * 0.007		= 0.11841		
	ϕ_{LT} = [py + (η_{LT} + 1) * pE] / 2		= 687.222 MPa		
Bending buckling strength	pb = pE * py / [ϕ_{LT} + SQRT(ϕ_{LT}^2 - pE * py)]		= 301.621 MPa	< py (355 MPa)	
Buckling resistance moment	Mb = pb * Zx = 436 kNm	>= m_{LT} *Mx = 69.7 kNm		**OK**	16%

TENSION + MOMENT CHECK

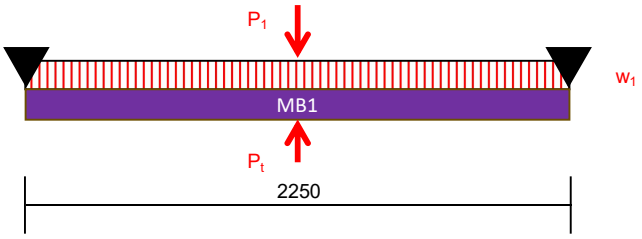
$$\frac{Nt}{Pt} + \frac{Mx}{Mcx} + \frac{My}{Mcy} = \frac{0}{4367} + \frac{69.7}{513} + \frac{0}{170} = 0.14 <= 1$$ **OK**

COMPRESSION + MOMENT CHECK

$$\frac{Nc}{A * py} + \frac{Mx}{Mcx} + \frac{My}{Mcy} = \frac{0}{4367} + \frac{69.7}{513} + \frac{0}{170} = 0.14 <= 1$$ **OK**

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Check: MB1



UC 305x305x97kg/m
S355

Loading

P_t = Test Load
= 1.4 x 67.5
= 94.50 kN

P_1 = Self-weight of concrete block on top supported by middle SB1
= 1.4 x 24 x 0.75 x 2.25 x 1.125
= 63.79 kN (factored)

w_1 = Self-weight of MB1
= 1.4 x 97 x 9.81/1000
= 1.33 kN/m (factored)

Design Shear, V = P/2 + wL/2
= (94.5-63.79) / 2 - 1.33 x 2.25 / 2
= 13.86 kN

Design Moment, M = $PL/4 + wL^2/8$
= (94.5-63.79) x 2.25 / 4 - 1.33 x 2.25^2 / 8
= 16.43 kNm

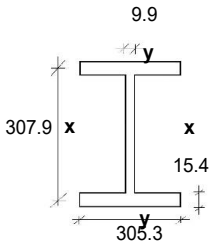
Structural Checking

(STRUCTURAL CHECKING REFER NEXT PAGES)

Calculation Sheet	Job No.	Sheet No.	Rev.
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	Member / Location:	MB1	
Job Tilte :	Drg. Ref.:		
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DESIGN CODE

Code of Practice for the Structural Use of Steel 2011



MEMBER SPECIFICATION

Member

Steel grade	=	BS S355	
Section	=	UC 305x305x97	(H-section)
Type	=	hot-rolled steel section	

MATERIAL/SECTIONAL PROPERTIES

Modulus of elasticity	E	=	205000	MPa									
Mass	m	=	96.9	kg/m									
Area	A	=	123	cm^2									
Dimensions	D	=	307.9	mm	d	=	246.7	mm					
	B	=	305.3	mm	b	=	152.65	mm					
	T	=	15.4	mm	t	=	9.9	mm					
Moment of inertia	Ix	=	22250	cm^4	Iy	=	7308	cm^4	Iu	= N.A.	cm^4	Iv	= N.A. cm^4
Radius of gyration	rx	=	13.4	cm	ry	=	7.69	cm	ru	= N.A.	cm	rv	= N.A. cm
Elastic modulus	Zx	=	1445	cm^3	Zy	=	479	cm^3					
Plastic modulus	Sx	=	1592	cm^3	Sy	=	726	cm^3					
	Svx	=	150.6	cm^3	Svy	=	717.7						
Torsional modulus	J	=	91.2	cm^4									
Design strength	py	=	355	MPa									

SECTION CLASSIFICATION

Parameter	ε	=	SQRT(275 / py)			=	0.8801	
Stress ratio	r1	=	Nc / (d * t * py)			=	0	
	r2	=	Nc / (A * py)			=	0	
Dimension ratio	b/T	=	9.9	<=	$\frac{15\,\varepsilon}{}$	=	13.2	Flange: Class 3
	d/t	=	24.9	<=	$\frac{80\,\varepsilon}{1+r1}$	=	70.4	Web: Class 1
								Overall: Class 3

DESIGN FORCES AND MOMENTS

Nc (kN)	Nt (kN)	Vx (kN) [along]	Vy (kN) [along]	Mx (kNm) [abt]	My (kNm) [abt]	Tr (kNm)
0.00	0.00	0.00	13.86	16.43	0.00	0.00

SHEAR CAPACITY CHECK

Shear area	Avy	= t * D	=	3048	mm^2					*(Cl. 8.2.1)*	
Shear capacity	Vcy	= py * Avy / SQRT(3)	=	624.76	kN	>=	Vy =	13.86	kN	OK	2%
Shear load check	Vy	<= 0.6 * Vcy	=	374.86	kN		(Low shear case)				
Shear area	Avx	= 0.9 * (2 * B * T)	=	8463	mm^2						
Shear capacity	Vcx	= py * Avx / SQRT(3)	=	1734.6	kN	>=	Vx =	0	kN	OK	0%
Shear load check	Vx	<= 0.6 * Vcx	=	1040.7	kN		(Low shear case)				

MOMENT CAPACITY CHECK

(Cl. 8.2.2)

(Low Shear Case)

Moment capacity	Mcx	= py * Zx	=	512.98	kNm	>=	Mx =	16.43	kNm	OK	3%
Moment capacity	Mcy	= py * Zy	=	170.05	kNm	>=	My =	0	kNm	OK	0%

BIAXIAL MOMENTS: LOCAL CAPACITY CHECK

$$\frac{Mx}{Mcx} + \frac{My}{Mcy} = \frac{16.43}{512.98} + \frac{0}{170.045} = 0.03 \quad <= 1$$

OK 3%

SHEAR STRESS AND TORSIONAL STRESS COMBINE CHECK

$$\frac{Vy}{Vcy} + \frac{Tr * T}{J*py} = \frac{13.86}{624.76} + \frac{0.00}{0.324} = 0.02 \quad <= 1$$

OK 2%

Calculation Sheet	Job No.	Sheet No.	Rev.
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	Made By:	Date: Oct-26	Check By:

TENSION CAPACITY CHECK

Effective Area $a_e = Ke * a_n$ = 1.1 x(12300 (Ag) - 0 (Area deduction for hole in mm²)) *(Cl. 8.6.1)*

= 13530 mm² >= 12300 mm² (Therefore take Ag)

Connection Type: **Welded Connection**

$a_2 = Ag - a_1$ = 9251.8 mm²

Tension capacity $Pt = py * (a_e - 0 \; a_2)$ = 4366.5 kN >= Nt = 0 kN **OK** 0%

COMPRESSION CAPACITY CHECK *(Cl. 8.7.5)*

Effective length	Lcx	= 2.25 m	Lcy	= 2.25 m	
Slenderness	λx	= Lcx / rx = 16.791	λy	= Lcy / ry = 29.2588	
From Table 8.7, use buckling curve		[b]		[c]	
Robertson constant	α	= 3.5	α	= 5.5	
	pEx	= $(\pi^2 * E) / \lambda x^2$ = 7176	pEy	= $(\pi^2 * E) / \lambda y^2$ = 2363	
Limiting slenderness	λo	= 0.2 * SQRT(π^2 * E / py) = 15.0988			
Perry factor	η	= α * (λx - λ0) / 1000 = 0.006	η	= α * (λy - λ0) / 1000 = 0.078	
	ϕ_C	= [py + (η + 1) * pEx] / 2 = 3787	ϕ_C	= [py + (η + 1) * pEy] / 2 = 1451	
Compressive buckling strength	pcx	= pEx * py / [ϕ_C + SQRT(ϕ_C^2 - pEx * py)] = 352.802 MPa	pcy	= pEy * py / [ϕ_C + SQRT(ϕ_C^2 - pEy * py)] = 325.592 MPa	
Compression resistance	Pc	= A * pcy = 4004.8 kN	>= Nc = 0 kN		**OK** 0%

LATERAL-TORSIONAL BUCKLING CHECK *(Cl. 8.3.1)*

Effective length	L_{LT}	= 2.25 m	m_{LT} = 1		
Slenderness	λ	= L_{LT} / ry	= 29.2588		
Dist. betw. flange shear centers	hs	= D - T	= 292.5 mm		
Web depth	hw	= D - 2 * T	= 277.1 mm		
wraping constant for channel	H	=	= 1.9E+12 mm⁶		
Torsional index	x	= 0.566 * hs * SQRT(A / J)	= 19.2264		
	γ	= 1 - Iy / Ix	= 0.67155		
Buckling parameter	u	= $(4 * Sx^2 * \gamma)/(A^2 * hs^2)]^{0.25}$	= 0.85161		
Slenderness factor	v	= $1 / [1 + 0.05 * (\lambda / x)^2]^{0.25}$	= 0.97298		
	βw	= Zx / Sx	= 0.90766		
Equivalent slenderness	λ_{LT}	= u * v * λ * SQRT(βw)	= 23.0974		
	pE	= $(\pi^2 * E) / \lambda_{LT}^2$	= 3792.51 MPa		
Limiting equivalent slenderness	λ_{L0}	= 0.4 * SQRT(π^2 * E / py)	= 30.1976		
Perry coefficient	η_{LT}	= (λ_{LT} - λ_{L0}) * 0.007	= -0.0497		
	ϕ_{LT}	= [py + (η_{LT} + 1) * pE] / 2	= 1979.51 MPa		
Bending buckling strength	pb	= pE * py / [ϕ_{LT} + SQRT(ϕ_{LT}^2 - pE * py)]	= 375.728 MPa	>= py (355 MPa)	
Buckling resistance moment	Mb	= pb * Zx = 513 kNm	>= m_{LT} *Mx = 16.43 kNm		**OK** 3%

TENSION + MOMENT CHECK

$$\frac{Nt}{Pt} + \frac{Mx}{Mcx} + \frac{My}{Mcy} = \frac{0}{4367} + \frac{16.43}{513} + \frac{0}{170} = 0.03 \;<=\; 1$$ **OK**

COMPRESSION + MOMENT CHECK

$$\frac{Nc}{A * py} + \frac{Mx}{Mcx} + \frac{My}{Mcy} = \frac{0}{4367} + \frac{16.43}{513} + \frac{0}{170} = 0.03 \;<=\; 1$$ **OK**

Web buckling checking (for unstiffened web)

Member **MB1 & SB1**

Section size

Compression force F_x = 94.5 kN

Stiff bearing length b_1 = t + 1.6r + 2T

= 65.02 mm

Section: D = 307.9 mm r = 15.2 mm

B = 305.3 mm

t = 9.9 mm

T = 15.4 mm

d = 246.7 mm p_{yw} = 355 N/mm^2

Check for web-to-flange connection

Connection location: Near ends of member b_e = 0 m

k = 30.6 mm (Rolled section) n = min(2 + 0.6b_e / k , 5)

= 2.00

Bearing capacity P_{bw} of unstiffened web

= (b_1 + nk) t p_{yw}

= 444 kN > F_x **OK**

Check for buckling resistance of web

Loaded flange is restrained against a) rotation relative to web

b) lateral movement relative to other flange: No

Distance from load to end of member a_e = 0 mm < 0.7d

Buckling resistance of unstiffened web P_x

= $25\varepsilon t P_{bw} / \sqrt{(b_1 + nk)\, d}$ x (a_e + 0.7d) / 1.4d

= 274 kN > F_x **OK**

Reduced buckling resistance of unstiffened web P_{xr}

= 0.7d P_x / L_E Effective length of web L_E = 277.1 m

= 171 kN > F_x **OK**

→ Therefore, stiffener in web is not requied.